

# SCHWA TO /A/: DEVELOPMENT OF UNSTRESSED VOWELS IN L2 SPANISH

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## ABSTRACT

Native (L1) English second language (L2) Spanish learners acquiring the Spanish vowel system are tasked with adjusting entrenched vowel categories and morphophonological rules. For example, allophonic variation of stressed and unstressed vowels in English is not present in Spanish, but the triggering environments for the variation is present in Spanish. As such, learners of Spanish must come to ignore those triggering environments during perception and production. The present study examines the production of unstressed Spanish /a/ by 10 L1 English L2 Spanish learners during a seven-week domestic immersion program in a semi-longitudinal study. Preliminary acoustic analysis revealed that within the first week, participants fronted and lowered their production of unstressed /a/, assumed here to correspond to less centralization and more target-like production. The findings are situated within current models of second language speech learning and provide often-sought longitudinal data on an understudied phenomenon.

**Keywords:** phonetics, phonology, second language acquisition, Spanish, English

## 1. INTRODUCTION

Adult native English (L1) learners of Spanish as a second language (L2) face difficulties producing target-like Spanish vowels due to a morphophonological centralization process in English in which unstressed vowels tend to reduce categorically to a centralized vowel. For example, a learner may produce the Spanish word /'ka.sa/ as non-target-like ['ka.sə]. The present study investigates the production of Spanish stressed and unstressed /a/ in 10 L1 English L2 Spanish beginner learners during a seven-week domestic immersion program in a semi-longitudinal design. Instead of extracting formant values at solely the midpoint of the vowel, acoustic analysis includes a measure of vowel trajectory and spectral centroid to obtain a more holistic view of vowel centraliza-

tion. Results provide information about typical development of L1 English L2 Spanish beginner learners with growing L2 exposure and how learners adjust their production in the face of growing evidence against a process present only in the L1.

## 2. LITERATURE REVIEW

### 2.1. Spanish and English Vowels

Vowel production has been found to be relatively uniform across Spanish varieties [1, 2], whereas descriptions of English vowel systems vary more widely [3, 4]. Spanish can be said to have a subset of English vowel phonemes /i e a o u/, but the phonetic realizations of these phonemes are not identical [4]. A major difference to highlight between the two systems is the impact of prosodic effects, such as lexical stress, on the realization of vowels. One of the major acoustic correlates of lexical stress cross-linguistically is vowel reduction in unstressed syllables [5]. Spanish has been found to have minimal centralization of vowels in unstressed syllables [6, 1], only representing a possibly universal low-level phonetic phenomenon [7]. On the other hand, English presents both low-level phonetic reduction like Spanish, as well as a language-specific, morphophonologic, categorical, rule-based process in which vowels in lexically unstressed syllables are realized as centralized vowels [ɪ] or [ə] [8, 9, 10]. For example, the first vowel in the English word 'atom' is realized as [æ] in the tonic syllable, but when realized in a non-tonic syllable like in the derived word 'atomic', the vowel is obligatorily reduced to [ə].

### 2.2. L1-L2 Transfer Effects

L1 English L2 Spanish learners typically do not present major issues acquiring phonological vowel contrasts [11], but the phonetic and morphophonological differences between their L1 and L2 may result in non-target-like production [12, 13, 14, 15, 16]. The acquisition path of Spanish vowels by L1 English L2 Spanish learners can be accurately described by

a number of frameworks such as the Speech Learning Model revised [17], the Perceptual Assimilation Model [18], or, as will be explored here, the Second Language Linguistic Perception (L2LP) model [19, 20, 21, 22, 23].

The L2LP model proposes that L2 learners create a new grammar via *full copying/full access*. Accordingly, difficulties in acquiring L2 speech sounds are accounted for by phonetic similarities, differences, or perceived equivalences to contrasts present in the L1. The learner must adjust their perception grammar to account for L2 input. Based on this framework, it is expected that linguistic processes present in the L1 may undesirably surface in the L2, but that these transfer effects may be reduced with increasing exposure to and proficiency in the L2. For example, [24] found that L1 English L2 Spanish late learners over a seven-week home-immersion program shifted perception and production of voice onset time categorical boundaries from initially English-like towards more Spanish-like values.

As mentioned, English presents vowel allophony in lexically stressed and unstressed positions not present in Spanish. In a cross-sectional study, [14] found that L1 English L2 Spanish late learners of Spanish, divided into “intermediate” and “advanced” proficiency groups, produced unstressed Spanish vowels as more centralized than L1 Spanish controls. Although valuable, [20] urges for the inclusion of longitudinal data from beginning language learners. Longitudinal L2 acquisition data has provided evidence, for example, that the formation of L2 phonetic categories can occur abruptly at early stages [25].

### 3. THE PRESENT STUDY

Previous research has pointed to the difficulties of L1 English L2 Spanish target vowel production, particularly when it comes to unstressed vowels due to L1-L2 transfer effects. Available studies typically present cross-sectional designs. From an L2LP perspective, cross-sectional designs only tell part of the story, and longitudinal data will reveal how the L2 grammar is progressively adjusted as the learner receives more input. As such, the present study tracks the production of /a/ in stressed and unstressed position in 10 L1 English L2 Spanish beginner late learners during a seven-week domestic immersion program in a semi-longitudinal design. With this data, the following research questions (RQs) are answered:

1. Do L1 English L2 Spanish learners centralize unstressed Spanish /a/?
2. Do L1 English L2 Spanish learners reduce centralization of unstressed Spanish /a/ as they

progress through a 7-week domestic immersion program?

Based on the L2LP [19, 20, 21, 22, 23], it was predicted that learners will initially produce unstressed /a/ as more schwa-like due to lexical stress effects present in the L1 [8, 13, 14], but that centralization would reduce over the seven-week period [24]. Furthermore, as has been seen in research on other segmentals [25], it is hypothesized that participants will demonstrate abrupt, as opposed to gradual, development in target-like production of unstressed /a/.

## 4. METHODOLOGY

### 4.1. Participants

Ten L1 English L2 Spanish late learners (6 females; ages 18–34 years,  $\bar{x} = 23.7$ ) were recruited for the present study. They reported no prior experience with any foreign languages, nor had they spent a significant amount of time in a foreign country. Participants were enrolled in the Middlebury Spanish Language School program’s beginner classes, placed via a placement test and interviews with two faculty members of the program. The Middlebury Language School program is a seven-week domestic immersion program where students are encouraged and quasi-obligated to use their target language at all times, resulting in high L2 input (from native and non-native speakers) and minimal L1 use.

### 4.2. Materials

Participants completed a delayed repetition task during the first week, and in subsequent weeks, a reading task. The analyzed materials consisted of 18 real and nonce words. Each word was a bisyllabic paroxytone with /a/ as the final vowel. While the first syllable variably contained an onset, the second syllable always contained an onset (e.g., *ada*, *ita*, *gaka*, *gota*). Auditory stimuli for the first session were produced in a carrier phrase by a 29 year old L1 Spanish speaker from Cádiz, Spain. Participants were recorded in a quiet classroom on site at Middlebury College using a Shure SM10A dynamic head-mounted microphone. A Sound Devices MM-1 pre-amplifier boosted the signal and sent it to a laptop computer where it was recorded using Praat at a 44.1 kHz sample rate with a 16-bit quantization [26].

### 4.3. Procedure

Participants completed the first experimental session on the second or third day of the program; further experimental sessions took place every Sunday, for a

total of 7 sessions. Stimuli were presented randomly to participants using PsychoPy2 [27], aurally for the first session and visually in subsequent sessions. Participants listened or read aloud each item 3 times. Each participant provided the dataset with 378 items for a total of 3780 tokens (18 words \* 3 repetitions \* 7 weeks \* 10 participants).

#### 4.4. Acoustic Analysis

Recordings were transcribed, segmented automatically [28], and manually corrected when necessary. A Praat script [?] was then used to extract formant values at 10% intervals of target vowels. A “spectral centroid” was calculated by averaging formant values at equidistant points spanning 20% to 80% of the vowel, which present less coarticulatory effects resulting from consonant environment [29, 30]. The trajectory length (TL) of vowels was also calculated, which represents a measure of formant movement [?]. Longer TL is associated with greater formant movement: “diphthongal” vowels tend to have TLs that exceed 620 Hz, whereas “diphthongized” vowels fall between 420 and 620 Hz [31].

#### 4.5. Statistical Analysis

Generalized additive mixed models (GAMMs) [32, ?] were fit to F1 and F2 centroids, measured in Hertz (Hz), with a parametric term for the variable *stress* to model mean differences in Hz between stressed and unstressed /a/. Smooth terms for *session* and *stress* by *session* were included to model non-linear effects of time for stressed and unstressed vowels. Another GAMM was fit to trajectory length (TL) with identical parametric and smooth terms. All three models included a random factor smooth by subject over session was included to model by-subject differences.

Models were interpreted using quantitative and visual methods [32]. *P*-values for the parametric *stress* term was evaluated at  $\alpha = 0.05$  to determine if there is a difference in overall mean Hz for stressed and unstressed vowels for F1, F2, and TL. Visual inspection of the difference plots were also used for the same comparison. Confidence intervals that do not overlap with 0 were taken as evidence that the two estimates are different. Lastly, F1, F2, and TL smooths were plotted to view their trajectories over session for a more individual, fine-grained inspection.

### 5. RESULTS

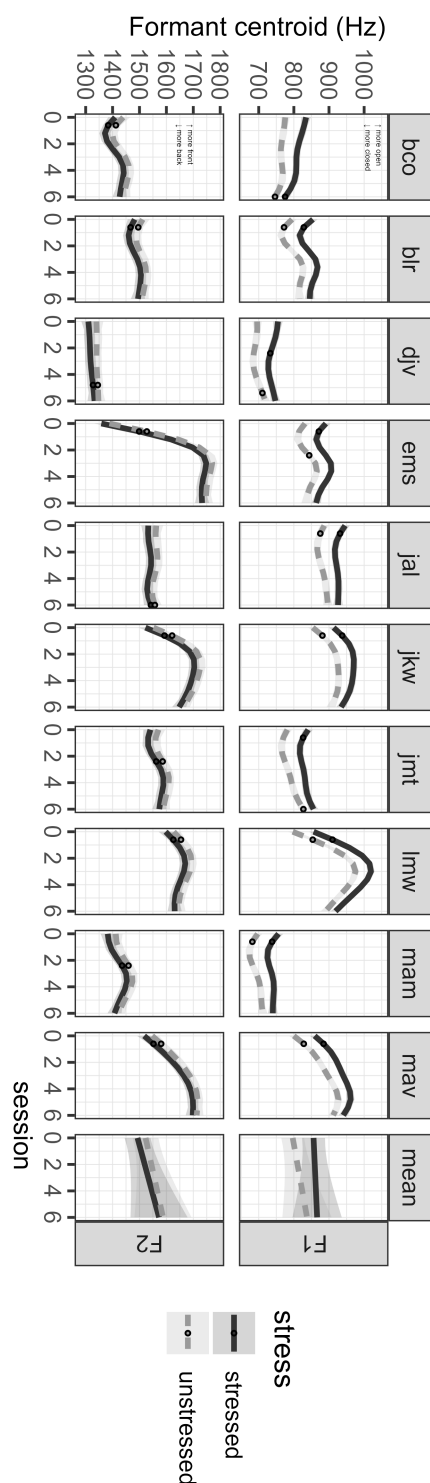
For F1, F2, and TL, stress was a significant parametric predictor. Unstressed /a/ had lower F1 ( $\beta = -44.07$ ,  $SE = 1.69$ ), higher F2 ( $\beta = 22.34$ ,  $SE = 3.33$ ),

and longer TL ( $\beta = 194.67$ ,  $SE = 13.17$ ). None of the population-level smooths reached significance for F1, F2, or TL. In contrast, participant-specific session smooths were significant in all three models (F1: EDF = 43.93,  $F = 206.26$ ; F2: EDF = 42.1,  $F = 122.8$ ; TL: EDF = 36.85,  $F = 66.93$ ), indicating substantial individual differences in trajectories for F1, F2, and TL across the seven sessions.

Visual inspection of the population means (Fig. 1) revealed that the learners were “closing the gap” between stressed and unstressed /a/ for F1 and F2, but difference plots (not printed here for lack of space) revealed that the formant differences between stressed and unstressed /a/ remained different from zero. For F1, stressed /a/ did not change much over the weeks, but unstressed /a/ clearly rose over the weeks to approach values similar to stressed /a/, indicating that unstressed /a/ production became more open. For F2, unstressed and stressed /a/ values were similar from the beginning and presented similar upward movement, indicating that /a/ production, regardless of stress, became more fronted.

### 6. DISCUSSION

The present work examined the production of /a/ in L1 English L2 Spanish beginner learners in a domestic immersion program in a semi-longitudinal design. Based on L2LP [19, 20, 21, 22, 23], it was expected that these learners would present centralization of unstressed /a/ due to L1-L2 transfer [8, 13, 14], but transfer effects would be mitigated with increasing L2 proficiency and exposure. In line with previous work, it was found that stressed and unstressed /a/ production differed in F1 and F2 at all time points. This is not particularly surprising, as even L1 Spanish controls and “advanced” learners presented slightly centered productions of unstressed /a/ compared to stressed /a/, although the two categories heavily overlapped for the L1 Spanish controls [14]. Importantly, F1 values for stressed and unstressed /a/ became more and more similar across the seven weeks, with unstressed /a/’s F1 raising while stressed /a/’s F1 maintained similar values, evidence for less centralization in unstressed syllables. The lack of movement for F1 in stressed /a/ is in line with previous work that indicates that F1 values for English and Spanish stressed /a/ in CVCV words are similar [4]. F2 values for stressed and unstressed /a/ both became higher across the seven weeks, indicating that the learners were fronting their /a/ more, in line with previous work identifying Spanish /a/ as more fronted than English /a/ [4]. Although the F2 difference plot showed a downward slope, there was no strong evidence that



**Figure 1:** Individual and population-level (right-most panel) F1 and F2 trajectories for stressed and unstressed vowels over 7 weeks. The moment of largest absolute change is marked with a circle.

F2 values were becoming more similar for stressed

and unstressed /a/.

Although /a/ productions were changed, there is no clear evidence from grand mean values that the changes were abrupt [25]. In fact, the GAMMs reports a linear, gradual change to F1 and F2 values across the weeks. Examining each participant's smooths 1, however, reveals that the moment of greatest absolute change in F1 or F2 for each stressed and unstressed /a/ occurred, in most cases, early on in the program.

## 7. LIMITATIONS AND FUTURE DIRECTION

There a number of improvements that can be made to the current study. L2 productions should be compared to a control group of bilingual speakers to determine what is specifically "on target". The current data analyzed only /a/, which may show a different development pattern than other vowels [14]. Trajectory length, for example, may prove to be a more useful measure of development for mid vowel production, which are typically diphthongized in English and transferred to Spanish. In terms of perception, the L2LP posits that perception precedes production [21], but the current data cannot provide evidence on this topic. In a case like this, perceptual boundaries of /a/ versus /ə/ could be investigated in L1 English L2 Spanish learners [33]. Lastly, accurate vowel categorization bears morphological weight in Spanish: centralized vowels may result in inaccurate perception or production of gender or verb morphology. As such, future investigations should examine vowel acquisition vis-a-vis morphological acquisition.

## 8. CONCLUSION

The present study analyzed the production of unstressed /a/ in 10 L1 English L2 Spanish beginner learners in a domestic immersion program in a semi-longitudinal design over seven weeks. Learners' stressed and unstressed /a/ productions became more similar across the seven weeks, and their F2 values became more target-like. This study deepens our understanding of how L1 morphophonological processes not present in the L2 may arise in L2 production and provides evidence for rapid development of fine-grained phonetics early in exposure to an L2.

### 8.1. References

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